

Predicting Final Listing Prices in a Private Used Car Market



– using Random Forest Algorithm

Group 2

Ji-Soo Kim, Tae-Yoon Kim, Jun-Beom Lee, Jin-Joo Yang, Sung-Hyun Kim

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1. Introduction

Car : Volkswagen - Golf

Bought : 2007

power : 245hp / Gasoline

...



1. Introduction

News

How much profit do car dealers make on new and used cars? Here's the perception versus reality

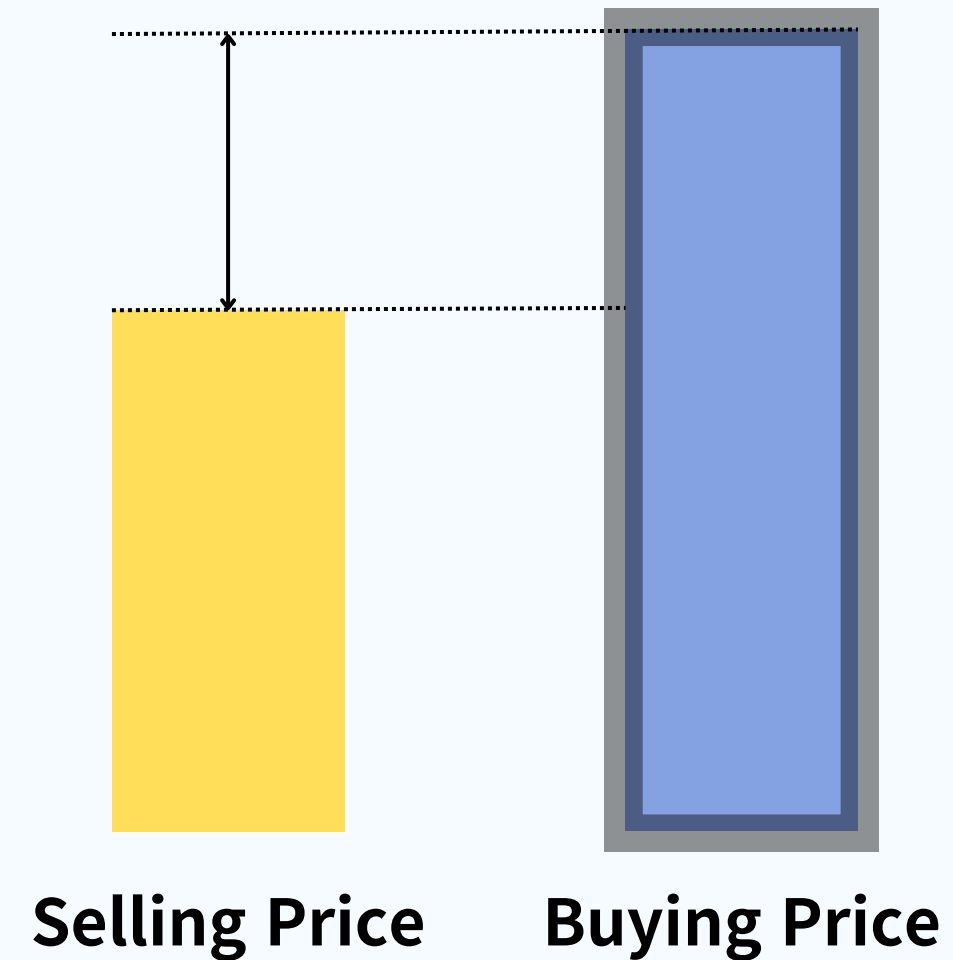
- Huge disparity between how much buyers think dealers make and the reality
- Some consumers think dealers are making more than 75 per cent profit per car
- What Car? survey shows average consumer thinks dealers make 10-20 per cent per unit
- Dealers reveal that they really make about seven per cent on new cars

The used car market is a lot stronger with profit margins for dealers around 12 to 15 per cent.

David Kendrick, partner and accountancy UHY Hacker Young, said the figures dealers quoted to Car Dealer are accurate.

He said: 'There is a huge misconception as to how much retailers make on vehicle sales.'

Dealers benefit from the difference between the selling price and the buying price of a used car



➔ **Sellers** want to maximize the selling price to obtain a greater profit, and still manage to sell the car

1. Introduction

BAIN & COMPANY Industries Consulting Services Digital Insights About Careers Explore

Brief

The Outlook for the European Used Car Market

Growth opportunities may be closer than they appear.

By Roch Baranowski, Eric Zayer, Klaus Stricker, and Ingo Stein
10 min read



With a total volume of 429 billion euros in 2021, the market for used cars in Europe is similar to the market for new cars. Of the 32 million used cars sold in Europe in 2021, some 44% were sold privately in consumer-to-consumer (C2C) transactions, while 56% were sold by professional retailers, which tend to focus on higher-class, younger cars. Furthermore, used car margins typically beat new car margins for dealers. Overall, the used car market is growing healthily, with a CAGR of 7% between 2015 and 2021.

Private Transactions are not easy to investigate with official statistics

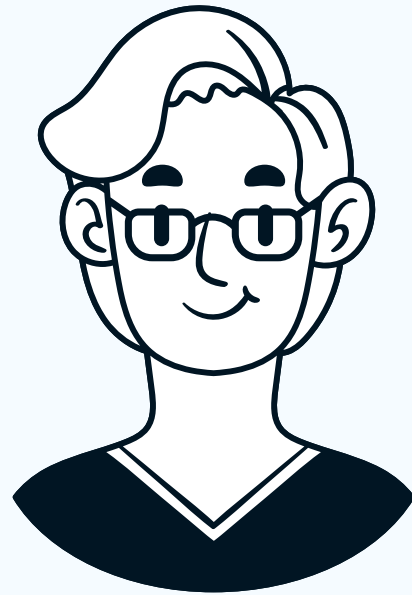
➔ **Buyers** struggle to determine if the listing price on the used car site is accurate or overpriced 🤔

1. Introduction

Let's Clarify the concept!

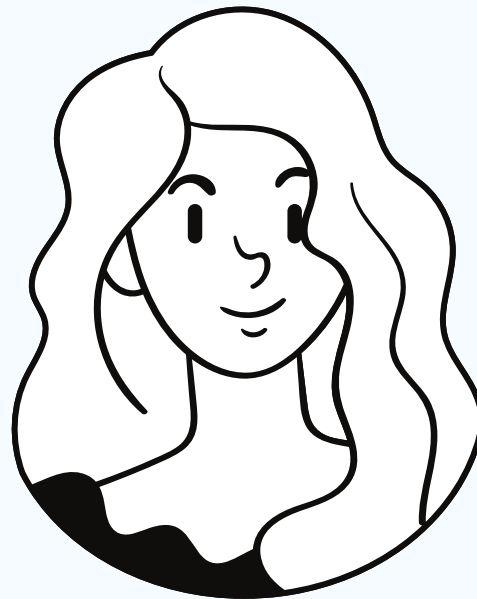
1. **Listing price** : the price that seller has set on the used car site
2. **Final Listing Price** : “Final registered price” after multiple adjustments
3. **Selling price** : the price that seller receives – the price that a seller sets in a used car market
4. **Buying price** : the total price customer ‘actually’ pays for the product (may include taxes and other fees)

2. Scenario Example - Target Audience



Daniel Brühl - Seller

- I'm trying to sell my original car in order to buy a new car!
- But, I'm afraid of the dealer fraud.
- Can't I just **sell this privately** at an **“Appropriate Price”**?



Maria Stein - Buyer

- I want to buy used car which is cheaper because I have a limited budget.
- I have to find a used car in private sale, without a dealer.
- How much is the **“common selling price”** of my dream car?

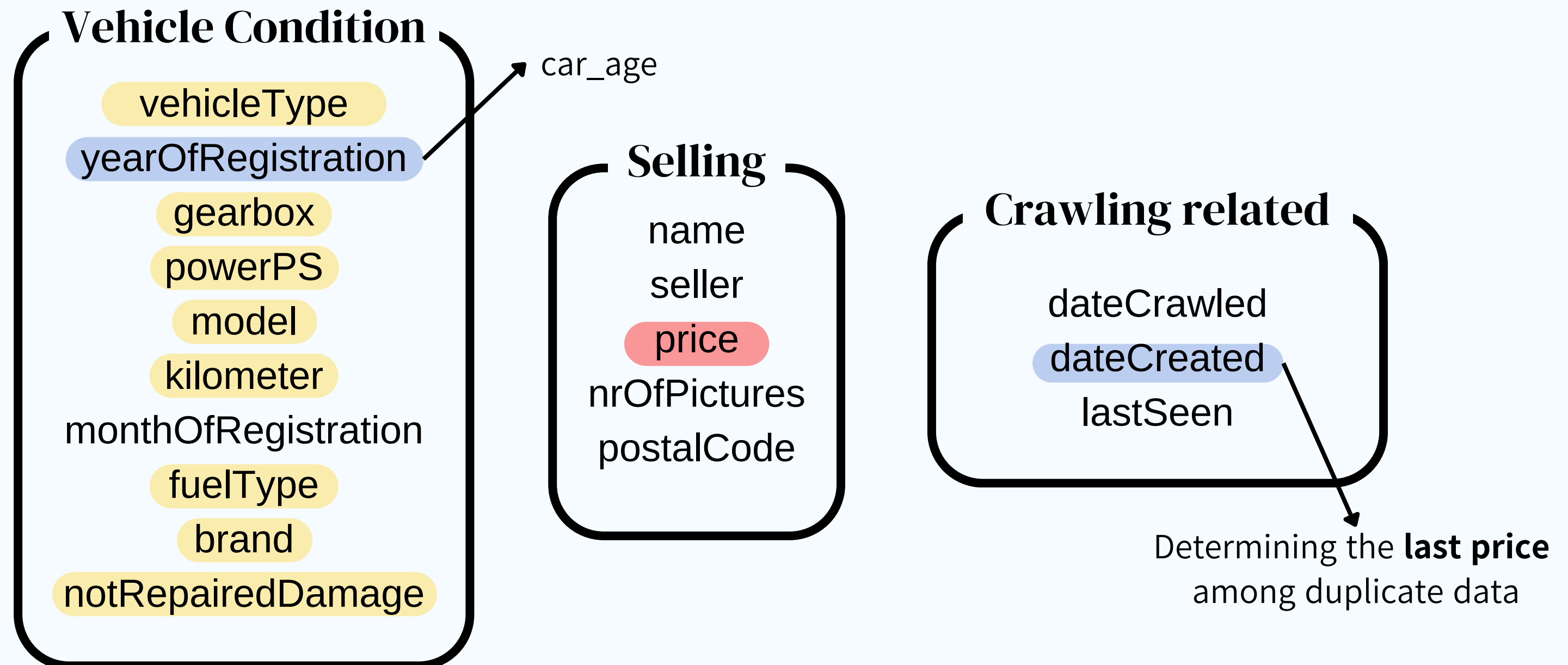
3. Our Dataset

Source of data : ebay or other sites where cars can be posted.

21 features

X

371,528 data samples



3. Our Dataset - Data Preprocessing

dateCreated	dateCrawled	lastSeen	name	price	vehicleType	yearOfR	gearbox	pow	model	kilomet	month	fuelType	brand
2016-04-04 0:00	2016-04-05 0:37	2016-04-07 3:17	***_ACHTUNG_FORD_KA_MIT_NEU_TÜV_BIS_04/2018_ZU_VERKAUFEN_***	1350	kleinwagen	2002	manuell	60	ka	150000	8	benzin	ford
2016-04-05 0:00	2016-04-05 8:49	2016-04-07 11:17	***_ACHTUNG_FORD_KA_MIT_NEU_TÜV_BIS_04/2018_ZU_VERKAUFEN_***	1350	kleinwagen	2002	manuell	60	ka	150000	8	benzin	ford
2016-04-05 0:00	2016-04-05 6:55	2016-04-07 10:44	***_ACHTUNG_FORD_KA_MIT_NEU_TÜV_BIS_04/2018_ZU_VERKAUFEN_***	1350	kleinwagen	2002	manuell	60	ka	150000	8	benzin	ford
2016-04-04 0:00	2016-04-04 22:54	2016-04-07 2:17	***_ACHTUNG_FORD_KA_MIT_NEU_TÜV_BIS_04/2018_ZU_VERKAUFEN_***	1350	kleinwagen	2002	manuell	60	ka	150000	8	benzin	ford
2016-04-04 0:00	2016-04-05 1:55	2016-04-07 10:16	***_ACHTUNG_FORD_KA_MIT_NEU_TÜV_BIS_04/2018_ZU_VERKAUFEN_***	1350	kleinwagen	2002	manuell	60	ka	150000	8	benzin	ford
2016-04-04 0:00	2016-04-04 22:50	2016-04-07 2:17	***_ACHTUNG_FORD_KA_MIT_NEU_TÜV_BIS_04/2018_ZU_VERKAUFEN_***	1350	kleinwagen	2002	manuell	60	ka	150000	8	benzin	ford

→
Data with the most **recent dateCreated** date
= **final listing price**

- Datasample having Null data that has been deleted.
- Duplications have been deleted if some of the feature's values were the same.
 - Range of price was limited from 500 to 245,000.
 - Range of powerPS was limited from 30 to 800.

4. Previous research analysis

Table 1: Performance Matrix of Linear Regression.

	Mixed Strategy	Label Strategy
Training MSE	43.73	54.26
Training R-squared	0.69	0.62
Test MSE	44.02	54.23
Test R-squared	0.70	0.63

Table 2: Performance Matrix of Decision Tree.

	Mixed Strategy	Label Strategy
Training MSE	0.02	0.02
Training R-squared	1.00	1.00
Test MSE	23.92	26.40
Test R-squared	0.84	0.82

J. He, "Predicting Vehicle Prices Using Machine Learning: A Case Study with Linear Regression,"
Applied and Computational Engineering, vol. 99, no. 1, pp. 35–42, Nov. 2024

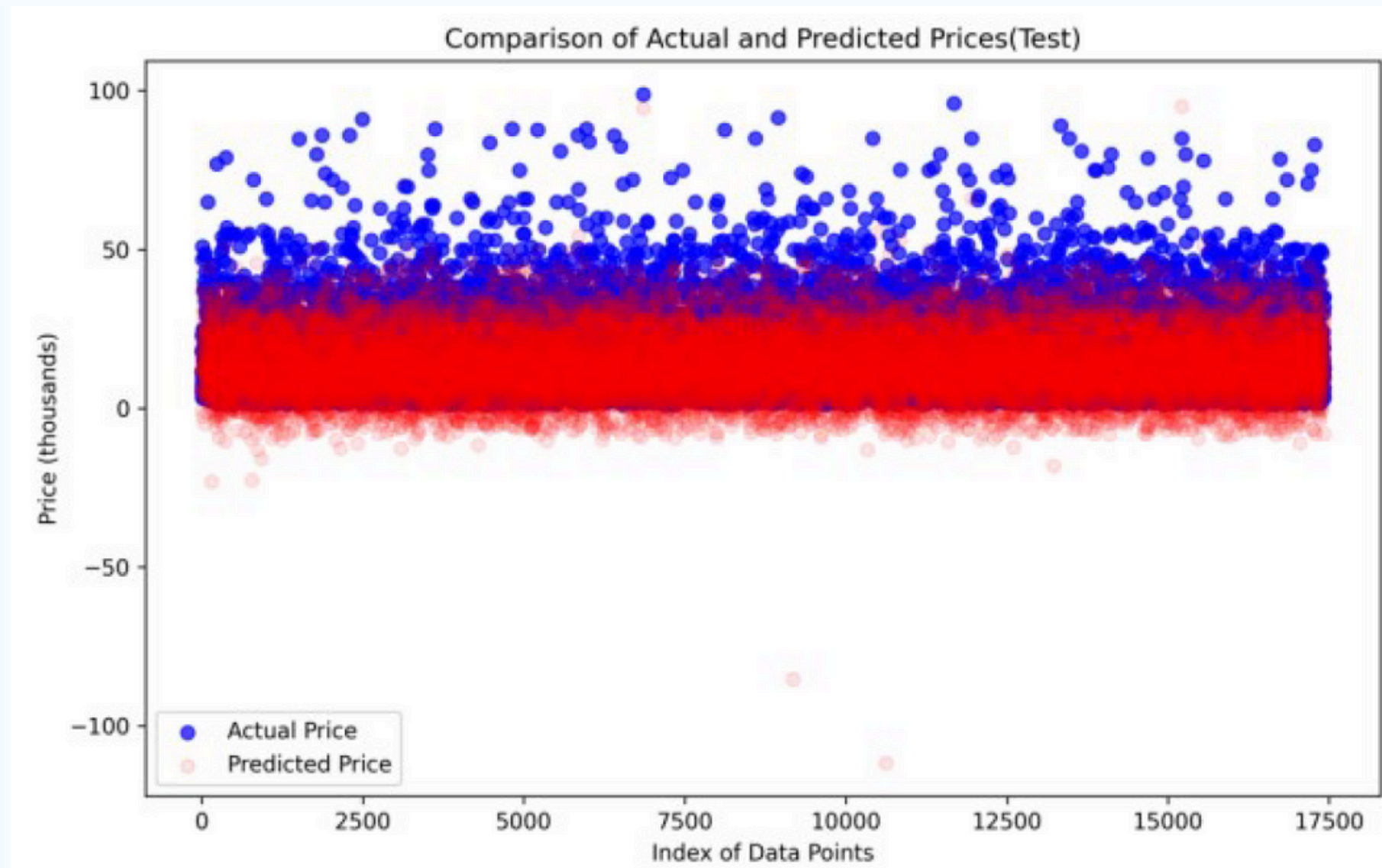
Overfitting -> random forest?

Mixed strategy - using one-hot encoding & manual mapping(label-encoding)

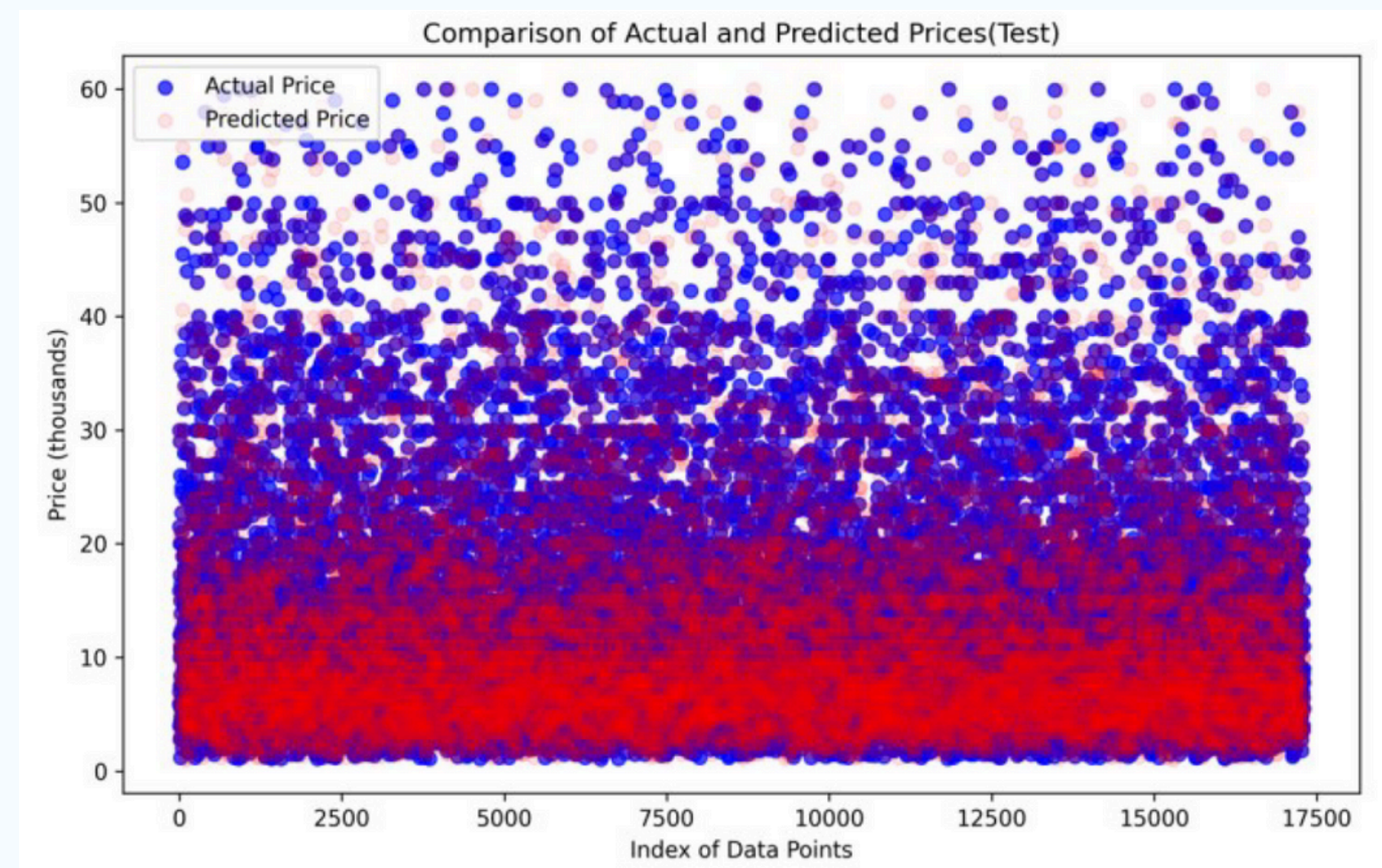
Label strategy - using only label-encoding

4. Previous research analysis

Distribution using linear regression



Distribution using decision tree



J. He, "Predicting Vehicle Prices Using Machine Learning: A Case Study with Linear Regression,"
Applied and Computational Engineering, vol. 99, no. 1, pp. 35-42, Nov. 2024

Low & medium prices : High Accuracy in both models

High price(luxury cars) : Accuracy declined

5. Model Works - Multiple Linear Regression (MLR)

Nominal values for MLR : Label encoding

Q. Why not One-hot Encoding?

A. Experienced the **Curse of Dimensionality**...🙄

Fixed) Linear correlation > 0.5 : car_age, kilometer, powerPS, car_model, car_brand

vehicle_Type(1), gearbox(2), fuel_Type(3), notRepairedDamage(4)

CASE	x	1, 3, 4	2, 3, 4	1, 2, 3, 4
R^2	0.63506842	0.648601815	0.656494454	0.655735274
mean absolute error	3174.356195	3132.950155	3118.916731	3124.028734
mean squared error	25281779.31	24440843.29	23588280.12	23578420.14
root mean squared error	5028.098976	4943.768127	4856.776721	4855.761541
mean signed difference	-5.4771E-10	-4.02865E-09	4.30216E-10	4.01539E-10
mean absolute percentage error	0.906901314	0.903172851	0.910689249	0.899603956
adjusted R^2	0.63506842	0.648601815	0.656494454	0.655735274

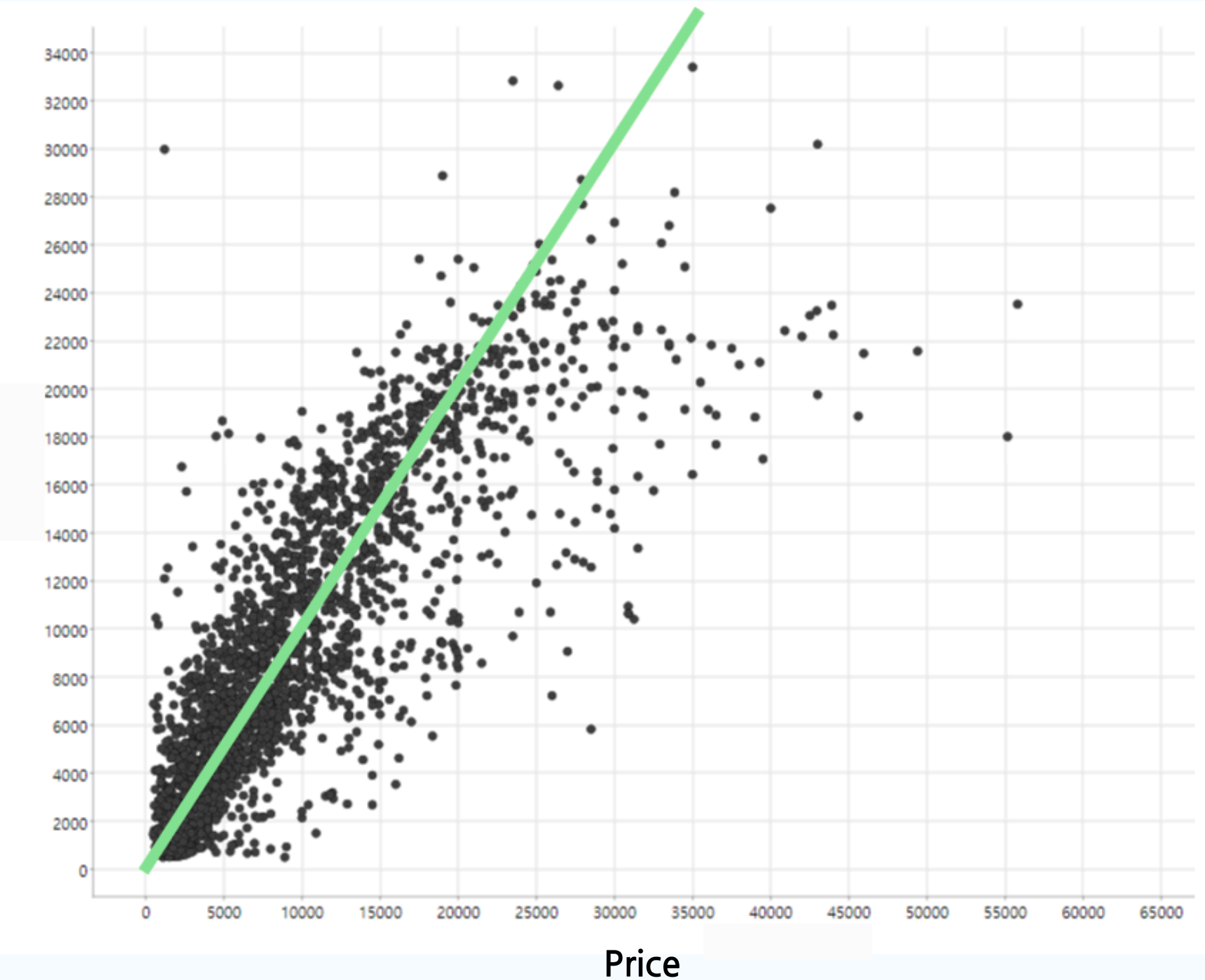
Using selected features

Using all features

5. Model Works - Multiple Linear Regression (MLR)

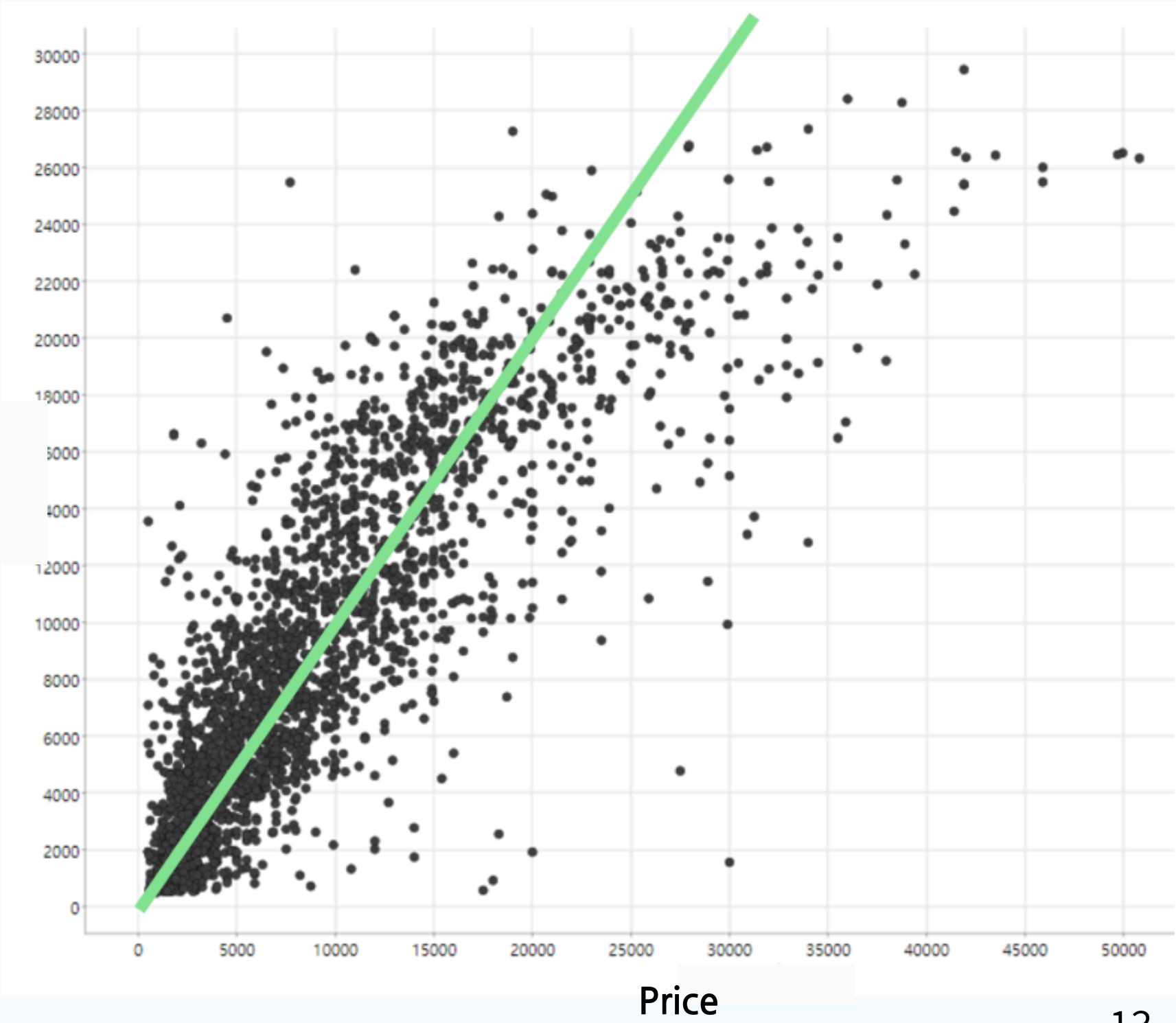
Prediction Price

Using selected features



Prediction Price

✓ Using all features



5. Model Works - Random Forest

Q1. Why Random Forest?

A. It's an ensemble model of individual decision trees, which is :

- Good at catching **nonlinear relationships**
- Good at handling **categorical variables** well
- Can effectively reflect interactions between variables
- Automatically calculates feature importance -> **use important variables**

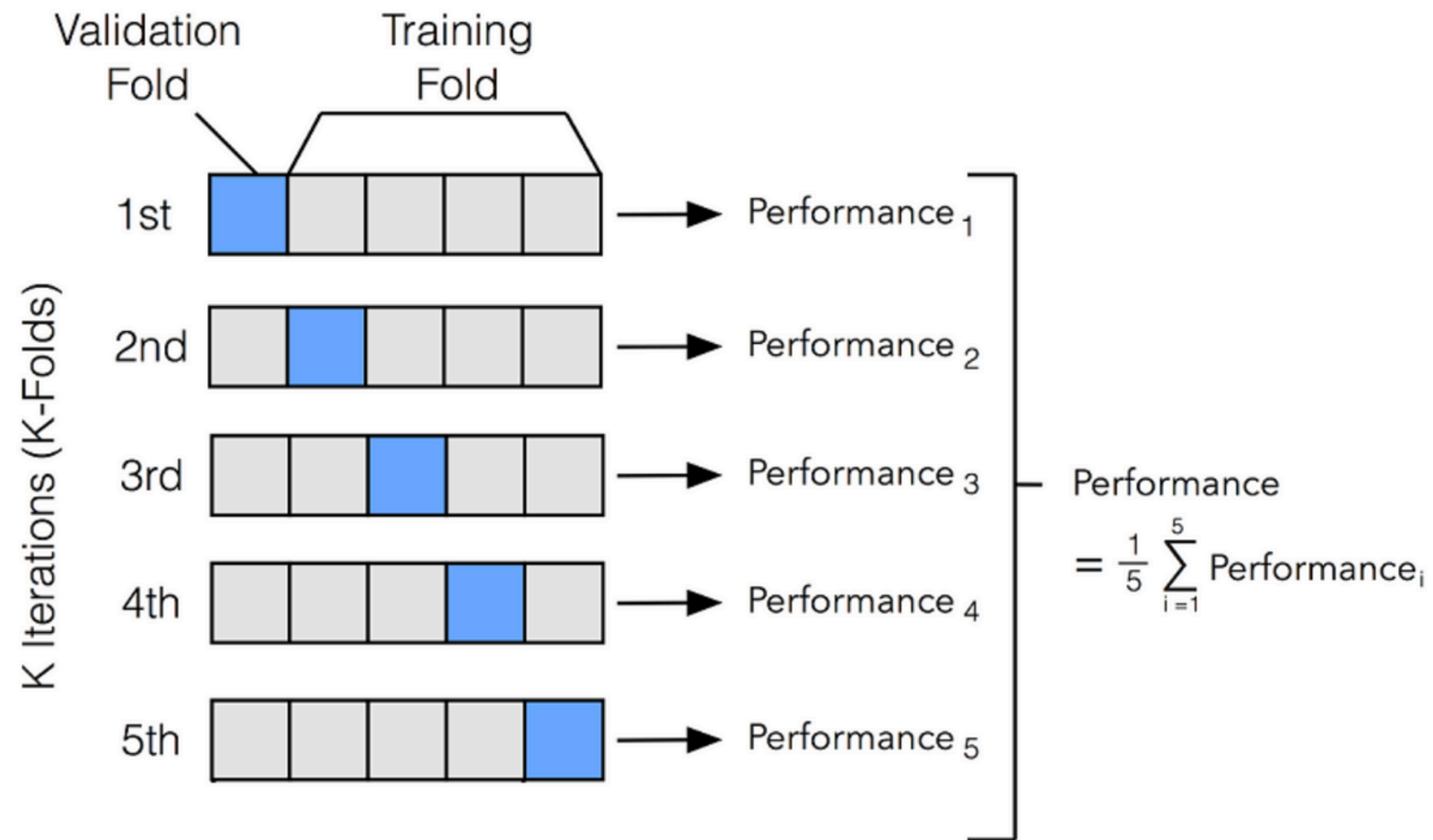
Q2. Why not MLP?

A. Without sufficient training data and adequate hyperparameter tuning, MLP may not capture nonlinearity well

- Requires **learning all its characteristics** & consideration of **non-critical variables** => Can lead to a lot of unnecessary learning!

5. Model Works - Random Forest

Random Forest Overfitting Prevent method:
K-fold validation



k-value: 5

$$80,000 / 5 = 16,000$$

Our dataset: about 80,000 data samples

➡ set the k-value 5 to adjust for the sufficient amount of data to train the model

5. Model Works - Random Forest

★Target Encoding..?

Convert brands and models to average prices

car_brand	car_model	car_brand_enc	car_model_enc	price
volkswagen	golf	6819.045426	6024.54069	600
volkswagen	golf	6819.045426	6024.54069	1459
volkswagen	passat	6819.045426	6013.214716	5890
volkswagen	golf	6819.045426	6024.54069	4000
volkswagen	touareg	6819.045426	17980.71884	11900
volkswagen	beetle	6819.045426	8735.380631	5800
volkswagen	fox	6819.045426	2997.835766	7800
volkswagen	golf	6819.045426	6024.54069	19200
volkswagen	golf	6819.045426	6024.54069	1200
volkswagen	golf	6819.045426	6024.54069	3799
volkswagen	golf	6819.045426	6024.54069	8990
volkswagen	golf	6819.045426	6024.54069	1750
volkswagen	passat	6819.045426	6013.214716	5100

5. Model Works Random Forest- UnEncoded

train

min child node size

tree depth	5	6	7	8	9	10	11	12	13	14	15
5	5073.83444	5087.78231	5085.24237	5044.77059	5051.80077	5036.44406	5022.2113	4940.41466	4976.87182	4931.20889	4969.2733
6	4647.40643	4637.04533	4648.1111	4645.80143	4610.97826	4596.37018	4576.86269	4579.94708	4568.39861	4547.30093	4551.12785
7	4286.65256	4291.06829	4299.23	4260.6506	4270.48117	4261.28607	4239.92579	4220.29771	4215.42981	4212.62722	4199.9638
8	3994.76992	3982.01599	3993.32079	3981.92781	3972.82208	3974.18785	3967.54015	3952.15442	3963.27206	3933.94239	3935.76502
9	3733.21848	3750.72231	3730.26372	3737.90289	3745.42784	3726.41265	3734.87151	3725.85411	3732.43839	3718.6797	3717.60498
10	3529.24836	3535.1556	3542.33743	3551.38359	3547.23559	3547.3813	3552.33057	3558.90025	3554.43746	3553.8165	3559.67442
11	3369.0179	3376.81598	3387.28097	3396.67537	3415.32879	3420.32519	3420.40687	3425.74612	3438.22344	3441.36099	3446.5621
12	3241.43168	3271.5179	3290.04411	3298.97172	3312.27157	3324.49011	3334.09417	3346.94211	3358.9221	3372.3531	3368.35463
13	3162.41366	3189.48057	3212.15548	3230.93479	3249.24921	3257.5106	3275.4621	3291.73784	3302.7549	3311.57033	3328.27638
14	3102.9982	3131.71174	3161.94235	3185.3263	3200.27528	3218.44648	3233.81834	3258.0882	3267.40402	3289.86268	3305.50835
15	3061.53513	3089.42134	3128.66421	3154.54102	3184.99155	3198.97043	3216.26569	3241.30883	3253.56628	3271.72241	3283.77644
16	3038.66982	3072.46229	3119.09173	3132.21421	3158.74277	3191.21771	3203.00865	3221.43159	3239.0058	3255.47362	3278.08264
17	3035.4788	3064.05504	3101.71589	3119.48314	3155.9946	3172.33532	3199.67726	3217.54133	3238.79261	3252.15388	3269.51797
18	3015.7388	3057.21754	3087.70831	3116.95178	3144.22538	3177.0075	3193.60756	3215.27044	3228.00326	3248.48157	3262.60705
19	3009.70148	3052.16202	3085.69582	3114.65252	3148.27473	3170.77217	3189.00195	3216.88805	3228.70332	3253.28522	3260.84319
20	3008.2095	3050.85801	3086.81845	3112.86636	3152.31314	3168.34755	3188.81149	3214.53585	3227.72573	3245.47777	3265.29016
21	3000.67154	3061.27253	3083.8327	3115.58255	3139.97258	3172.88853	3195.33989	3208.75505	3223.69141	3246.40279	3258.06007
22	3004.03584	3049.22275	3093.32643	3122.96191	3145.98031	3169.79873	3194.29973	3213.42431	3231.49244	3250.78816	3270.25276
23	3012.44444	3046.83626	3083.93522	3116.89608	3137.21002	3171.00817	3187.88489	3206.87074	3226.35983	3243.66767	3268.39063
24	3004.37493	3038.71688	3081.16252	3118.21506	3131.76292	3166.46486	3197.95679	3212.23677	3224.7554	3252.23578	3266.77799
25	3010.44305	3051.95382	3084.03992	3119.33657	3150.52851	3167.03154	3186.27009	3209.68453	3232.00133	3255.09312	3265.93106
26	3013.96995	3044.68477	3084.43694	3112.13656	3146.67059	3170.51559	3185.20054	3211.34698	3232.56138	3246.59824	3266.92676
27	3010.08922	3051.82265	3082.57837	3118.07079	3145.00023	3157.70852	3191.64799	3209.84602	3229.52443	3236.01902	3261.01255
28	3006.54145	3049.09129	3088.95345	3108.38729	3142.97688	3163.5209	3189.26531	3211.99551	3220.0202	3256.10877	3262.85876
29	3006.95744	3042.70042	3087.14048	3110.67243	3135.62192	3167.84852	3184.85411	3212.58475	3229.79439	3252.35345	3270.33652
30	3006.7741	3038.68252	3084.66253	3113.41696	3138.40477	3174.73642	3189.58678	3213.00669	3232.10213	3253.25515	3272.05503

test

		min child node size										
tree depth		5	6	7	8	9	10	11	12	13	14	15
Tree Depth	5	5093.47771	5109.62795	5106.24872	5063.58482	5071.26392	5054.3477	5043.28366	4963.52804	4998.25032	4950.50096	4991.23021
	6	4680.67791	4671.16165	4681.16953	4682.34446	4644.18101	4629.01604	4614.12403	4614.52255	4601.31609	4579.42669	4582.98891
	7	4337.76804	4344.19437	4354.50249	4311.49435	4320.8563	4316.31406	4286.92099	4270.53968	4266.54417	4261.78976	4252.52373
	8	4080.16097	4059.46884	4075.90732	4052.88733	4038.49405	4047.4435	4037.21588	4017.67734	4031.99517	4006.34606	4001.82316
	9	3837.02517	3859.02027	3838.91911	3835.71708	3840.61894	3817.44119	3822.23436	3807.10305	3816.35368	3806.80277	3807.53992
	10	3673.5571	3670.74591	3672.08289	3673.5668	3660.69495	3666.66953	3663.45594	3664.05112	3664.04457	3654.97334	3656.51459
	11	3534.26907	3535.05772	3543.1302	3543.93062	3551.09499	3558.6598	3545.59387	3548.77584	3556.45728	3561.73253	3555.82406
	12	3443.73527	3449.84617	3455.00377	3456.33513	3460.00625	3465.71269	3478.35871	3487.25453	3493.19795	3492.19636	3490.78558
	13	3373.40715	3391.27326	3392.86655	3401.365	3409.32018	3421.21469	3423.04055	3425.75864	3444.64247	3450.73978	3459.67467
	14	3331.43707	3334.0127	3355.88946	3368.89879	3375.54895	3387.56466	3387.04664	3408.5984	3413.14728	3426.03688	3436.77229
	15	3302.22637	3305.30342	3335.44474	3339.73756	3356.83001	3370.11938	3380.45864	3393.86067	3404.08069	3402.78334	3424.02275
	16	3279.70356	3304.50721	3318.08862	3329.75259	3340.48747	3356.58512	3372.46615	3371.88733	3389.26525	3397.24868	3413.02665
	17	3284.51907	3300.60914	3311.48173	3311.42345	3338.09722	3359.6985	3364.8549	3377.87255	3387.2519	3395.27921	3411.30409
	18	3263.78099	3281.73814	3300.01274	3319.49509	3329.35701	3351.6599	3366.37679	3377.16981	3380.09196	3391.502	3404.53193
	19	3264.46511	3287.39162	3303.72491	3311.02813	3337.71025	3353.58877	3350.53496	3376.65431	3382.56478	3395.14178	3400.77209
	20	3271.96764	3294.80225	3307.26351	3313.44267	3332.46753	3347.96015	3354.7165	3369.40298	3383.27171	3393.95814	3401.44157
	21	3270.48203	3299.8496	3300.03276	3312.96723	3330.9407	3348.24316	3354.99309	3364.59504	3374.62207	3398.13076	3391.29174
	22	3264.17155	3275.9566	3314.87641	3326.66326	3329.93637	3346.28239	3362.87239	3367.725	3384.5016	3398.0871	3406.78118
	23	3266.24493	3280.75883	3298.43235	3313.54974	3327.3832	3349.33532	3350.27213	3361.5134	3377.8521	3388.17196	3406.61434
	24	3270.49148	3275.45112	3304.90942	3317.39314	3322.71257	3343.4402	3363.02126	3370.66467	3375.03526	3396.30248	3409.56203
	25	3273.55845	3279.96193	3300.01667	3323.07215	3335.6864	3347.63704	3361.70353	3362.71317	3382.04699	3401.97518	3403.8082
	26	3270.1202	3279.12595	3297.97358	3316.11863	3332.73596	3348.71876	3354.97296	3369.31893	3387.51886	3401.34329	3407.22835
	27	3267.71373	3300.13122	3293.05863	3316.79336	3330.44077	3338.13372	3360.5385	3365.51531	3379.96571	3385.01038	3406.66746
	28	3257.10923	3291.48586	3306.7723	3308.36474	3329.28972	3341.4195	3357.96373	3369.85321	3380.10069	3398.77737	3402.7839
	29	3261.13332	3294.11766	3303.18545	3309.24626	3330.24269	3345.36613	3353.69786	3373.73174	3382.046	3397.29149	3411.1507
	30	3267.25544	3276.41766	3306.67593	3321.46739	3334.02764	3348.43867	3358.96828	3374.52829	3383.48752	3399.58586	3412.41225



Limit number of levels (tree depth)

Number of tree levels to be learned.

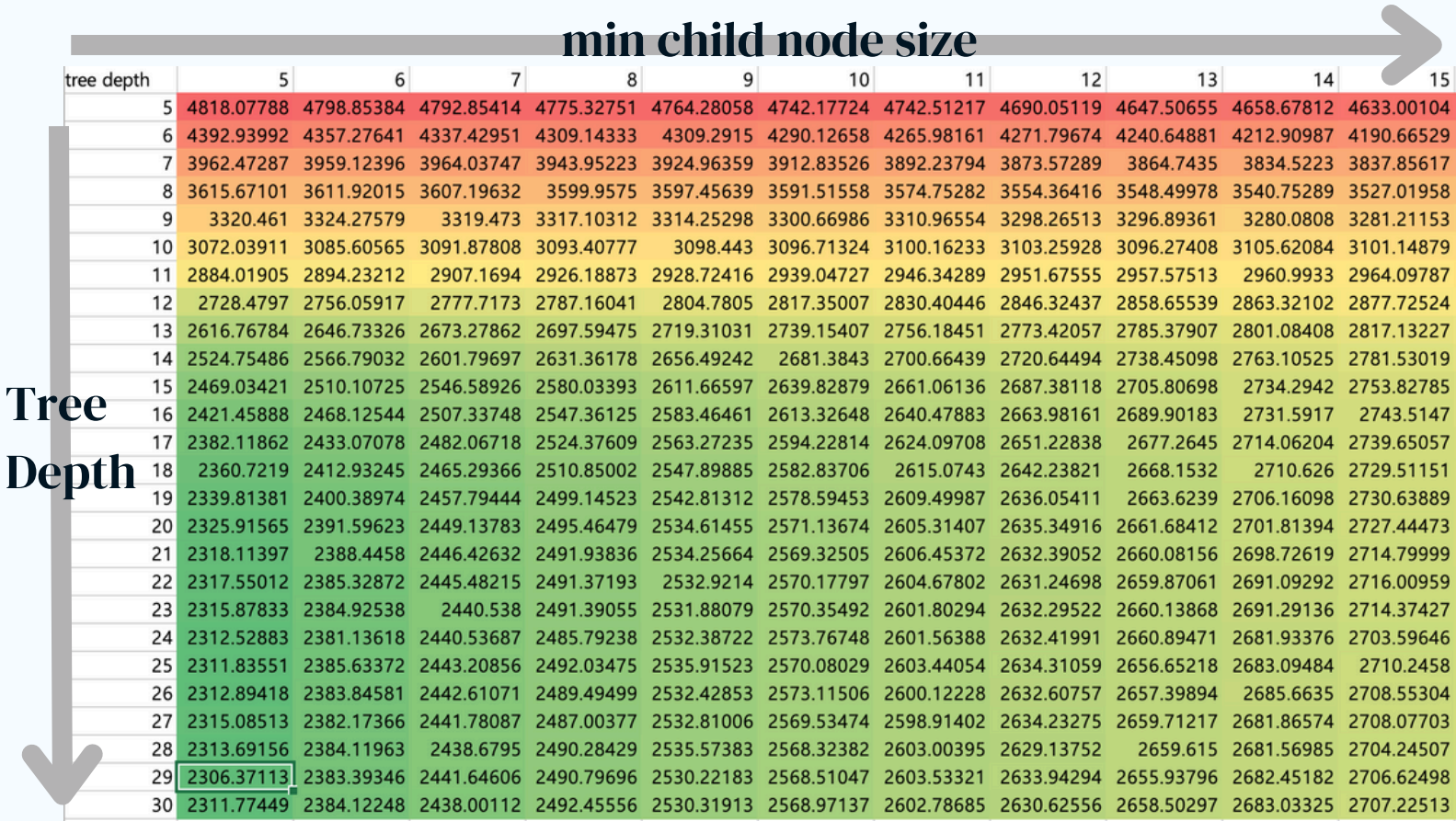
Minimum child node size

Minimum number of records in child nodes

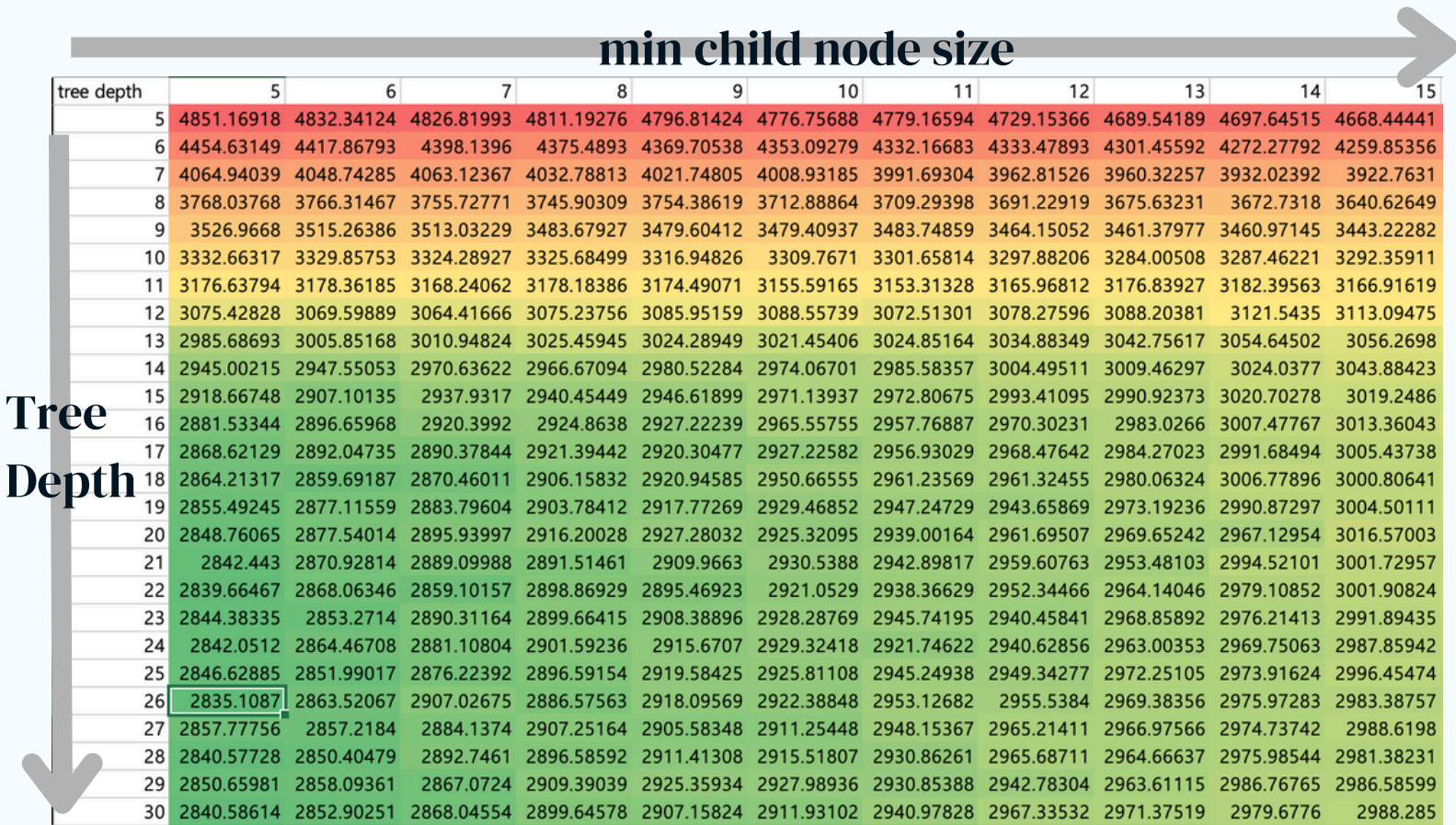
Limit number of levels (tree depth)	Minimum child node size
28	5

5. Model Works Random Forest- Encoded

train



test



Limit number of levels (tree depth)

Number of tree levels to be learned.

Minimum child node size

Minimum number of records in child nodes

Limit number of levels (tree depth)	Minimum child node size
26	5

6. Model Results

Original data (un-encoded)

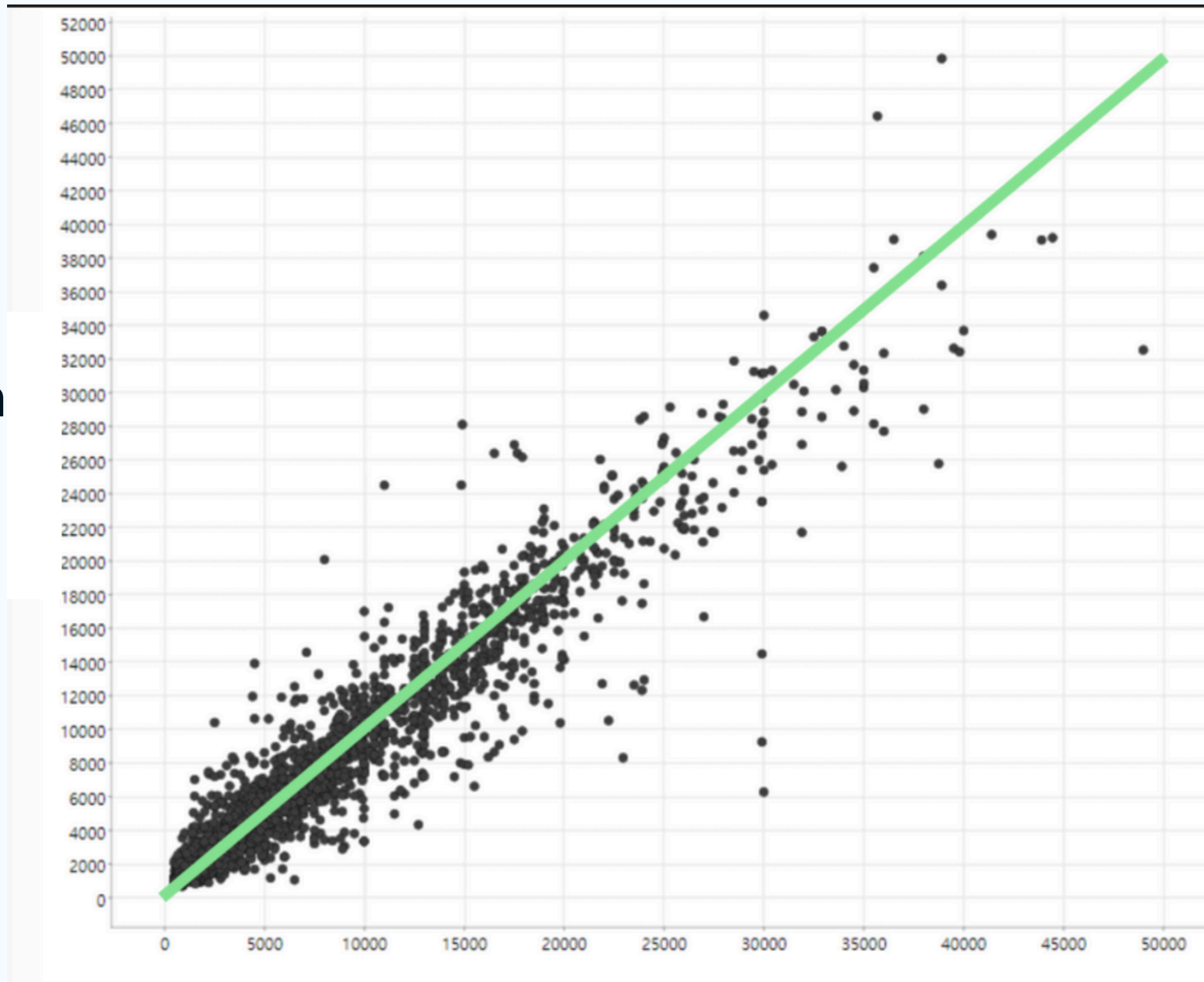
performance	prediction(price)
R ²	0.847
MAPE(Mean Absolute Percentage Error)	0.455
RMSE(Root Mean Squared Error) [€]	3257.109

Target Encoded data

performance	prediction(price)
R ²	0.885
MAPE(Mean Absolute Percentage Error)	0.319
RMSE(Root Mean Squared Error) [€]	2833.331

6. Model Results

Prediction
Price



Price

Q. How to resolve big errors in RMSE?

A. Dividing into Low price / High price

★Criterion?

6. Model Results

Low price

performance	prediction(price)
R ²	0.878
MAPE(Mean Absolute Percentage Error)	0.324
RMSE(Root Mean Squared Error) (€)	1732.82

High price

performance	prediction(price)
R ²	0.602
MAPE(Mean Absolute Percentage Error)	0.112
RMSE(Root Mean Squared Error) (€)	6190.55

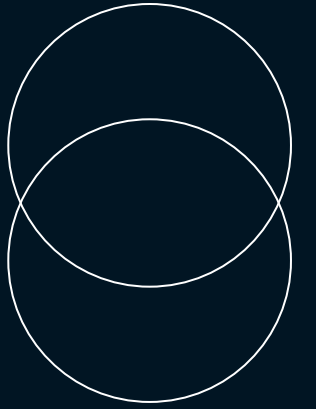
7. Conclusion

1. Based on the MLR result : Poor performance when the model was turned with label encoded data in MLR...
2. So we used the Random Forest Algorithm!
3. Even though R^2 was high and performed high accuracy, why is the RMSE too high?
4. So even though the MAPE size is similar, the RMSE gets bigger as the price gets higher. -> In Low/medium price, RMSE may be small

Low & medium prices : Predicts Well!

Luxury Cars : Seems to perform low..

5. Divided the low/medium price and the high price
-> it actually decreased from medium to low prices!



Q & A

Thank you for listening!